

Hisham Panamthodi Kajahussain

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Summary

Backend-focused Software Engineer with 4 years of experience building enterprise **Java/Spring Boot** microservices and scalable, event-driven, cloud-native **AWS** systems across healthcare and life sciences domains. Improved the performance and operational efficiency of enterprise backend systems through process automation, API optimization, and cloud-native architecture. Bring strong **backend** and **cloud-native** development expertise, with experience building **AI-powered applications**.

Education

California State University (CSUF), Fullerton, CA

Aug 2024 – May 2026

M.S. in Computer Science

GPA: 3.9/4

Relevant Coursework: Data Structures and Algorithms, Machine Learning(ML), Software Design and Architecture, Big Data

APJ Abdul Kalam Technological University (KTU), Kerala, India

Jul 2015 – May 2019

B.Tech in Computer Science and Engineering

GPA: 3.81/4

Professional Experience

Tata Consultancy Services (TCS)

Cochin, India

Software Engineer | Java, Spring Boot, Microservices, REST APIs, AWS, Kubernetes, PostgreSQL

July 2019 – July 2023

- Improved Bayer AG's campaign operational efficiency by **35%** by developing a **Java/Spring Boot** middleware application that automated end-to-end campaign workflows including campaign targeting, campaign simulation, target file generation, and **Salesforce Marketing Cloud** campaign delivery across **4 countries**.
- Implemented event-driven simulation pipeline using **RabbitMQ**, **Spring WebClient**, and **WebSocket** to decouple compute-intensive Sales Rep-to-HCP mapping from synchronous HTTP threads, reducing campaign simulation turnaround time by **60%** with real-time frontend notifications.
- Integrated **AWS Secrets Manager**, **S3**, and **IAM Assume Roles** to secure credentials, file storage, and cross-account access; contributed to **Dockerized ECS Blue-Green CI/CD** for **zero-downtime production releases**.
- Developed **PostgreSQL** data access layers using **Spring Data JPA** and **Hibernate** for multi-stage campaign persistence, and implemented **Spring AOP** audit logging to enforce end-to-end traceability, reducing root-cause analysis time by **30%**.
- Engineered **Java/Spring Boot** microservices for Bayer AG's Crop Science platform monitoring pesticide, herbicide, and fungicide resistance in agricultural field samples, supporting sample-to-report workflows for **10,000+ samples annually**.
- Decomposed a legacy Oracle-based monolith into **7 domain-driven microservices** on **AWS EKS**, enabling independent deployments and scaling; implemented a 7-state sample lifecycle and **Amazon SES** alerts for state changes.
- Reduced API response times by **45%** by introducing **Redis ElastiCache** TTL caching and optimizing **MySQL** reads for high-frequency reference data under concurrent load.
- Implemented enterprise API security using **Spring Security 5**, **JWT RS256**, and **AD/LDAP**, enforcing **4-tier RBAC** across **7 microservices** with centralized JWT validation at the API Gateway.

Skills

Languages & Frameworks: Java, Spring Boot, JPA/Hibernate, Python, FastAPI, JavaScript, React, Node.js, HTML/CSS

Cloud Platforms: AWS (EC2/S3/RDS/ECS/EKS/Lambda/IAM/SNS/Secrets Manager)

DevOps & Tools: Docker, Kubernetes, Terraform, Git/GitHub Actions, CI/CD, Kafka, RabbitMQ, JUnit/Mockito

Databases: MySQL, PostgreSQL, MongoDB (NoSQL), Redis, ChromaDB

Academic Projects

AI Clinical Decision Support System | GitHub

Jan 2026 – May 2026

- Built an end-to-end clinical AI platform using **Java/Spring Boot**, **4 Python/FastAPI microservices**, and **LLaMA 3.2** RAG to transform unstructured patient notes into ranked diagnoses, treatment recommendations, and explainable AI insights, enabling clinicians to receive actionable decision support from a single patient case submission.
- Fine-tuned **Bio_ClinicalBERT** on **24,445 clinical samples** achieving **87.45%** accuracy on **24-class ICU diagnosis**; deployed on **Amazon Web Services (AWS) ECS Fargate** via **Terraform IaC** and **GitHub Actions** Continuous Integration/Continuous Deployment (CI/CD), reducing manual deployment from hours to under 10 minutes.

CloudShelf E-Commerce Platform | GitHub

Aug 2025 – Dec 2025

- Built a cloud-native bookstore platform on **AWS EKS** with **9 independently deployable microservices** and polyglot persistence (PostgreSQL, MongoDB, Redis) across **3 AWS Availability Zones**, eliminating any single point of failure.
- Designed an async order notification pipeline using **Apache Kafka** and **AWS SNS**, fully decoupling checkout from email delivery so that order confirmation reaches users instantly regardless of downstream service availability.
- Integrated **Resilience4j** Circuit Breaker (50% failure threshold) to prevent cascading failures and automated the CI/CD pipeline for **9 microservices** via **GitHub Actions**, **Docker**, and **AWS ECR**, delivering zero-downtime rolling deployments.

Publications

- Hisham P. Kajahussain, *MedIntelliX – AI-Powered Medical Decision Support System*, **IEEE CHASE 2026**.